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Introduction to AI Agents on Azure

Learn to Develop, Test, and Deploy AI Agents with Microsoft Foundry

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Microsoft CERTIFIED Trainer

Microsoft CERTIFIED AZURE SOLUTIONS ARCHITECT EXPERT

Microsoft CERTIFIED AZURE AI ENGINEER ASSOCIATE

Microsoft CERTIFIED AZURE DATA SCIENTIST ASSOCIATE

Microsoft Azure Fundamentals (AZ-900) Certification Course, 2nd Edition

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With your instructor
Reza Salehi

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Associated roles

Cloud native engineer | Cloud solutions architect | Cybersecurity engineer | Database administrator

+1 more

Skills covered

AZ-900: Microsoft Azure Fundamentals
AZ-303: Microsoft Azure Architect...
AZ-500: Microsoft Azure Security...
AI-900: Microsoft Azure AI Fundamentals

Includes quizzes

Test your knowledge during the course and with a final quiz.

Beginner

Instructional content for a learner who has little or no knowledge on this topic

Learning Outcomes

- Gain knowledge of Azure cloud concepts and services
- Explore Azure services in greater depth
- Get ready for Exam AZ-900: Microsoft Azure Fundamentals
- Comfortably work with the Azure portal

The Microsoft Azure Fundamentals (AZ-900) exam is one of the most popular certifications for those who are just beginning to work with cloud-based solutions and services or who are new to Azure. The exam certifies knowledge of cloud concepts, Azure services, workloads, security and privacy, and pricing and support.

In this self-paced course, Reza Salehi will help you get familiar with Microsoft Azure's cloud services and begin your Azure certification journey. This course is aligned to the AZ-900 exam objective domains and has recently been updated to reflect the most current version of the exam (2024). It covers all the services and concepts in the Azure ecosystem you need to know in order to prepare for the test.

What you'll learn and how to apply it

By the end of this certification course, you will understand the following:

- General cloud concepts
- Core Azure services
- Core solutions and management tools on Azure
- General security and network security features
- Identity, governance, privacy, and compliance features
- Azure cost management and service-level agreements

Azure Cookbook

- https://learning.oreilly.com/library/view/azure-cookbook/9781098135782/
- https://www.amazon.ca/Azure-Cookbook-Recipes-Maintain-Solutions/dp/1098135792/
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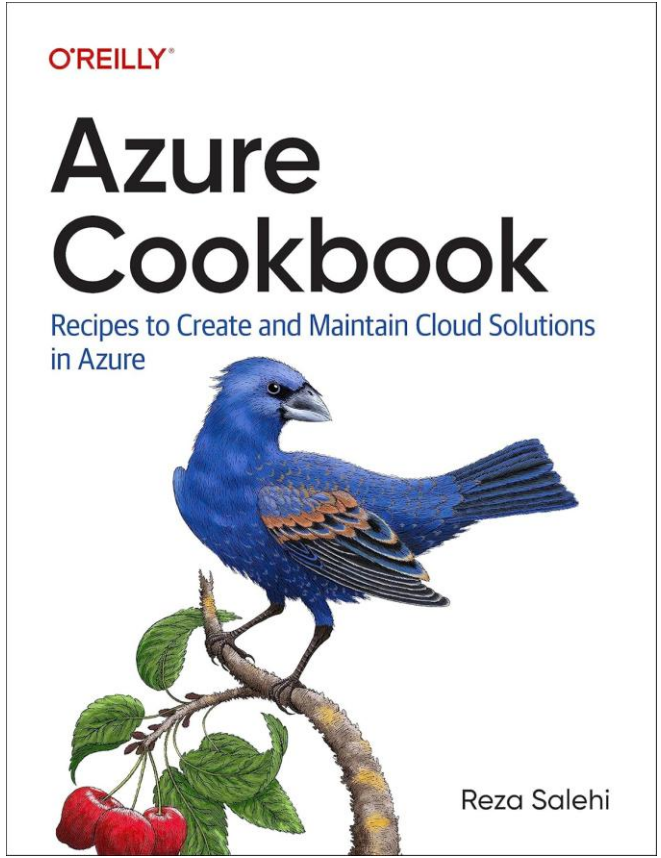


Course Overview

Learning Objectives

By the end of this course, you will:

- Gaining familiarity with Microsoft Foundry
- Use a variety of AI models for your AI agent
- Leverage Microsoft Foundry for agent development

Course Repository

https://github.com/zaalion/oreilly-ai-agents-designing

CERTIFICATION

Microsoft Certified: Azure AI Fundamentals

Demonstrate fundamental AI concepts related to the development of software and services of Microsoft Azure to create AI solutions.

At a glance

Level
Beginner

Product
Azure

Role

Subject

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate

Design and implement an Azure AI solution using Azure AI services, Azure AI Search, and Azure Open AI.

At a glance

Level
Intermediate

Product
Azure

Role

Subject

CERTIFICATION

Microsoft Certified: Agentic AI Business Solutions Architect

As an AI-first solution architect, you lead the transformation of enterprise operations by envisioning and implementing AI-powered architecture.

At a glance

Level
Advanced

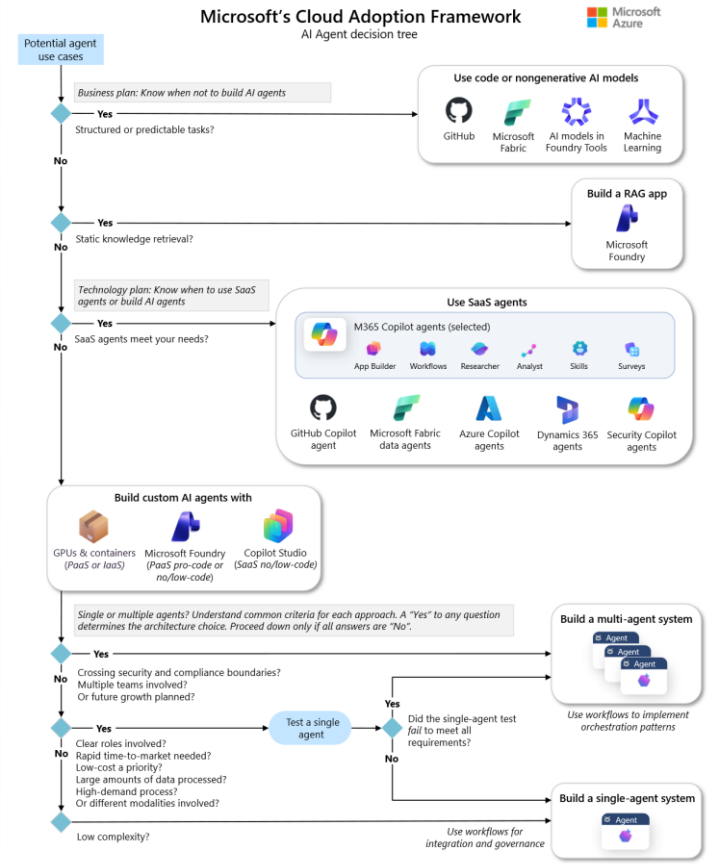
Product
AI Builder

Role
Solution Architect

Subject
Artificial intelligence

Renewal Frequency
12 months

Technology Plan For AI Agents



https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ai-agents/technology-solutions-plan-strategy

Pulse check: Have you developed an AI agent?

Understanding LLMs in Microsoft Foundry

Azure OpenAI in Azure AI Foundry Models

- Azure OpenAI Service
  - Available models
  - Pricing
  - Quotas & limits
  - Deploy, and use a model
  - Upgrade to Microsoft Foundry
- Microsoft Foundry (Azure AI Foundry)
  - Model catalog
  - Agent Playground
  - Instructions, actions, triggers, tools, RAG

Poll: What is Azure OpenAI Service?

- A marketplace for third-party AI apps
- New name for Azure Machine Learning
- A database service for storing AI data
- A managed service to access OpenAI models in Azure

Azure OpenAI in Azure AI Foundry

1. **Azure OpenAI**
    a. Provides REST API access to OpenAI's powerful language models including gpt-5 series, o4-mini, and more.
2. **Microsoft Foundry**
    a. Unifies agents, models, and tools under a single management grouping with tracing, monitoring, evaluating, and deployment capabilities.

Azure OpenAI Models

- Models available
  - gpt-5 series
  - o4-mini & o3
  - gpt-4.1
  - computer-use-preview
  - o3-mini & o1
  - o1-mini
  - GPT-4o & GPT-4o mini
  - GPT-4 series (including GPT-4 Turbo with Vision)
  - GPT-3.5-Turbo series

Azure OpenAI vs. OpenAI

- OpenAI + Azure = Azure OpenAI
  - Native integration with other Azure resources
  - Rich tooling (Azure AI Foundry, Visual Studio, VS Code, etc.)
- Azure OpenAI co-develops APIs and models with OpenAI
- Is compatible and provides smooth transition from OpenAI.

Azure OpenAI Pricing: Pay Per Token

GPT-5 series

Model	Pricing (1M Tokens)
GPT-5 2025-08-07 Global	Input: $1.25 Cached Input: $0.13 Output: $10
GPT-5 Data Zone	Input: $1.38 Cached Input: $0.14 Output: $11
GPT-5-mini Global	Input: $0.25 Cached Input: $0.03 Output: $2
GPT-5-mini Data Zone	Input: $0.28 Cached Input: $0.03 Output: $2.20

https://azure.microsoft.com/en-us/pricing/details/cognitive-services/openai-service/

Azure OpenAI Pricing: Pay Per Token

Microsoft | Azure Explore Products Solutions Pricing Partners Resources Search Learn Support Contact Sales Get started with Azure Sign in

Pricing calculator

Calculate your estimated hourly or monthly costs for using Azure.

Get started with Azure

Log in to save cost estimates and use your Azure agreement pricing.

Products | Example scenarios | Saved estimates | FAQs

Select a product to include it in your estimate.

openai

Azure OpenAI

Apply advanced coding and language models to a variety of use cases

Add to estimate

https://azure.microsoft.com/en-us/pricing/calculator

Azure OpenAI Pricing: Token Types

- Tokens
  - Text tokens
    - Azure OpenAI processes text by breaking it into tokens.
    - Can be words or chunks of characters (ham.bur.ger, pear, etc.)
  - Image input tokens
    - Level of image detail (low or high) + the image dimensions
  - Image generation tokens
    - Image dimensions and quality

https://learn.microsoft.com/en-us/azure/ai-foundry/openai/overview#tokens

Poll: What is Microsoft Foundry?

- A tool for labeling AI training data
- A platform to build, customize, and deploy generative AI apps
- A service for hosting machine learning notebooks
- A marketplace for pre-trained AI models

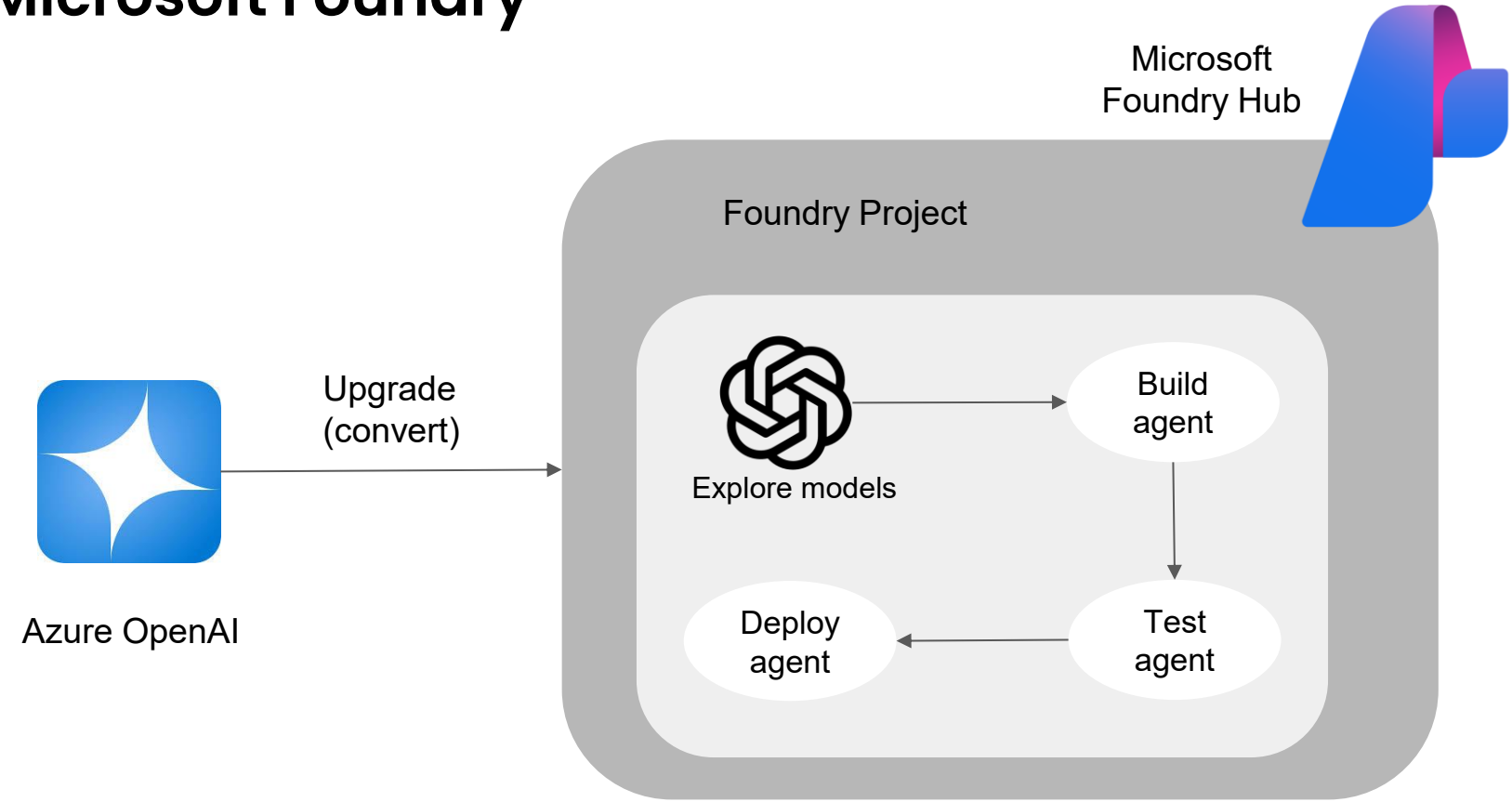
Azure AI Foundry

- As a developer, Microsoft Foundry helps you:
  - Explore, build, test, and deploy generative AI models
    - Grounded in responsible AI practices.
  - Build generative AI applications and AI agents
  - Collaborate with your team for the full development life-cycle
  - Work across model providers with a consistent API interface
    - OpenAI and other 3rd party models

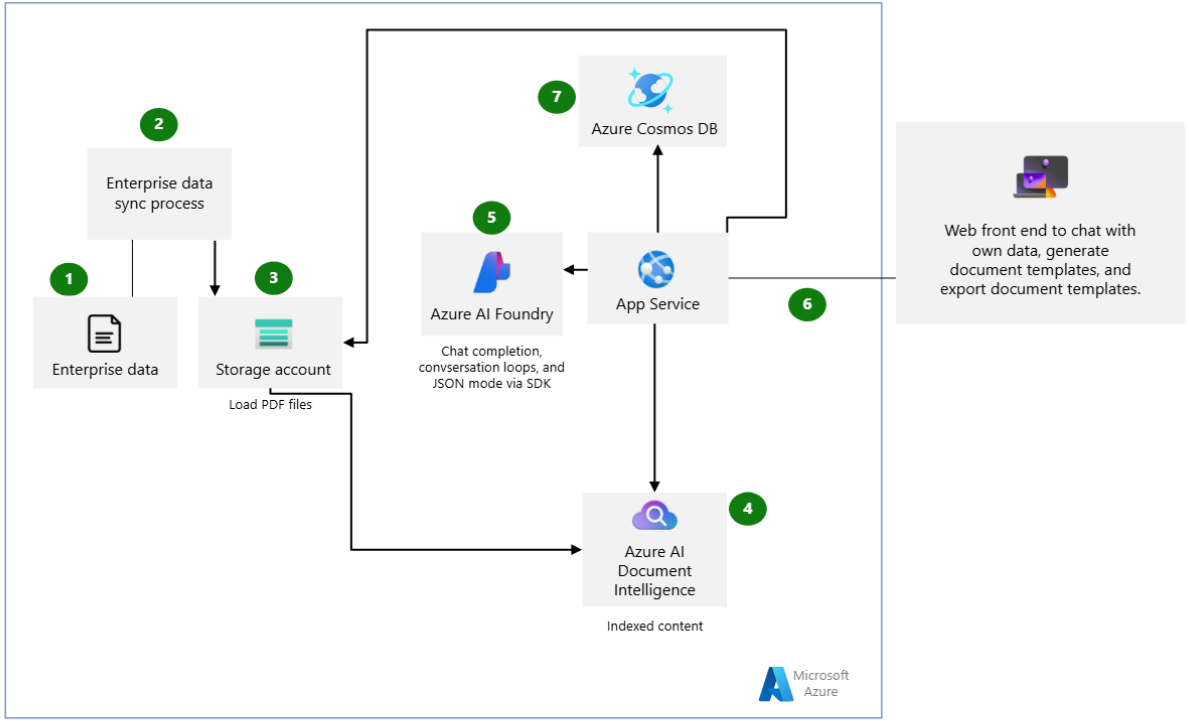
Microsoft Foundry

- Microsoft Foundry API and SDKs
  - Python
  - C#
  - JavaScript/TypeScript
  - Java

Microsoft Foundry



Microsoft Foundry



https://learn.microsoft.com/en-us/azure/architecture/ai-ml/idea/generate-documents-from-your-data

Microsoft Azure

Upgrade from Azure OpenAI to Microsoft Foundry

- For new projects, start with Foundry
- For existing Azure OpenAI resources, consider upgrading:
  - Microsoft Foundry offers more capabilities compared to the Azure OpenAI
  - Access to a broader model catalog, agents service, and testing capabilities
  - An upgrade option is available
  - Keep existing Azure OpenAI API endpoint, state of work, and security configurations
  - No need to create a new Foundry resource

Azure OpenAI in Microsoft Foundry Models Quotas and Limits

1. Regional quota allocation
2. Quotas and limits reference

Microsoft Foundry Playgrounds

- Foundry playgrounds:

  a. AgentOps support for Evaluations and Tracing in the **Agents playground**

  b. **Images Playground 2.0** for gpt-image-1, Stable Diffusion 3.5 Large, and Bria 2.3 Fast models.

  c. **Audio playground** for gpt-4o-audio, gpt-4o-transcribe, and gpt-4o-mini-tts models.

  d. **Video playground** for Azure OpenAI Sora.

  e. Open in **VS Code** for **Chat and Agents playground**. This feature saves you time by automatically importing your endpoint and key from Foundry to VS Code for multi-lingual code samples.

Demo:

Azure OpenAI

Demo:

Azure OpenAI Upgrade to Microsoft Foundry

Demo:
Azure OpenAI in Microsoft Foundry

Demo:
Add and Configure Models to Microsoft Foundry

Demo:
Microsoft Foundry Chat Playground

Introduction to AI Agent Development in Microsoft Foundry

Introduction to AI Agent Development in Microsoft Foundry

- Understanding Foundry Agent Service
  - Exploring the Agent Playground
- Creating a new agent
  - Choose the right LLM model
  - Triggers
  - Instructions
  - RAG
  - Tools
  - Threads
  - Actions

Poll: Which generative AI models can't be used by Foundry Agent Service?

- Azure OpenAI LLM models
- All Azure OpenAI models
- Azure OpenAI and other 3rd party models such as DeepSeek
- Azure AI Services models

Developing Agents with Azure AI Foundry

- **Foundry Agent Service**
  a. Connects models, tools, and frameworks into the agent.
  b. From generative model demos to a production-ready product:
     i. Deploying a LLM model
     ii. Create and test a new agent
     iii. Deploy to production
     iv. Monitor, tune, and improve

Developing Agents with Azure AI Foundry

- **AI Agent**
    a. Is not a chatbot!
    b. Makes decisions, invokes tools, and participates in workflows
        i. Independently
        ii. Collaboration
            1. Other agents
            2. Humans
    c. Enable AI-powered process automations

Poll: Your Azure AI Agent is using a model trained with data up to September 1, 2023. Can it provide answers about events that happened after that date?

- Yes, but it will hallucinate
- Yes, if you implement RAG
- Yes, it has real-time internet access
- No, unless you upgrade the subscription

Retrieval Augmented Generation (RAG)

1. AI model is the brain of the agent
    a. Has a cut-off date (e.g. January 15th, 2023)
2. RAG enables your agent to retrieve up-to-date data from various sources:
    a. Sharepoint
    b. Files
    c. Azure AI Search (+ underlying databases)
    d. Bing Search (web search)
    e. …

Developing Agents with Microsoft Foundry

- **AI Agent components**

  a. **Model (LLM):** Powers reasoning and language understanding

     i. Azure OpenAI models

     ii. Other 3rd party models

  b. **Instructions:** Define the agent's goals, behavior, and constraints

  c. **Tools:** Let the agent

     i. Retrieve knowledge (RAG)

     ii. Take action (send emails, etc.)

  d. **Trigger**

     a. Activates an AI agent to start working

Developing Agents with Microsoft Foundry

- Microsoft Foundry is the assembly line for AI agents, following these steps:

  a. Select a **model**, as the agent's "brain"

  b. Customize the model to fit your project

  c. Use **AI tools** to retrieve knowledge (RAG) or take an action

  d. Use connected agents for **collaboration/orchestration** scenarios

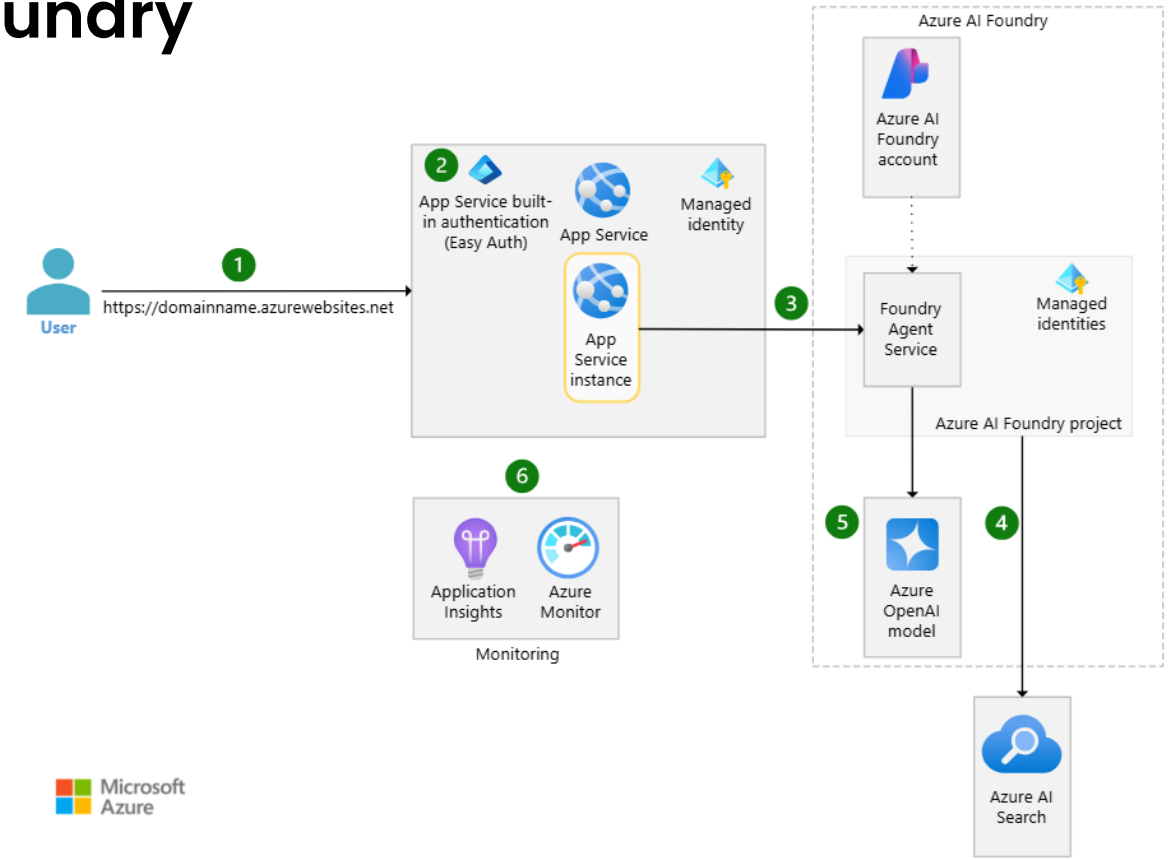
  e. Test and monitor your agent

  f. Foundry applies enterprise-grade **security** features including Entra ID support, RBAC, content filters, encryption, and network isolation

Tools in Microsoft Foundry Agent Service

- **Agent tools**
  a. Enables your AI agent to retrieve data or to take actions and automate workflows
  b. Tool types:
     i. Knowledge tools
        1. Private or public data: Azure AI Search, File Search, Microsoft Fabric, Bing Search, etc.
     ii. Action tools
        1. Azure Logic Apps, Azure Functions, etc.

Microsoft Foundry



https://learn.microsoft.com/en-us/azure/architecture/ai-ml/architecture/basic-azure-ai-foundry-chat

Microsoft Azure

Demo:
Create Azure AI Agent Project in Foundry

Demo:
Microsoft Foundry Agents Playground

Demo:
Develop, and Test Agents

Demo:
Deploy Your AI Agent

Introduction to Coding AI Agents

Introduction to Coding AI Agents

- Use AI Foundry to generate code for model calls, agents
- Introduction to Semantic Kernel
- Introduction to AutoGen
- Introduction to Microsoft Agent Framework

Poll: Which tool is currently in public preview?

- Microsoft Foundry
- Microsoft Agent Framework
- AutoGen
- Semantic Kernel

Use AI Foundry to Generate Code for Model Calls, Agent

- **Microsoft Foundry**
  a. Generates code snippets
  b. Lets you build AI agents and integrate the latest AI models into your code
     i. C#
     ii. Python
     iii. REST API
     iv. ...

Developing Agents with Semantic Kernel

- **Semantic Kernel**
  a. Single agents, tool/plugin orchestration, enterprise integration
  b. Develop agents in other platforms, other than Azure
  c. Build AI agents and integrate the latest AI models into your code
     i. C#
     ii. Python
     iii. Java
  d. Can add your existing code as a plugin

Developing Agents with AutoGen

- **AutoGen**
  a. Multi-agent system, complex conversations, experimentation
  b. Develop agents in other platforms, other than Azure
  c. Open-Source Framework for Agentic AI
  d. QuickStart

Microsoft Agent Framework (in preview)

a. **Microsoft Agent Framework** is a framework for building AI agents
    a. That can reason, use tools, access data, and act autonomously
    b. Standardizes agent design, orchestration, and tool use
    c. Why?
        a. Microsoft: "*It combines AutoGen's simple abstractions for single and multi-agent patterns with Semantic Kernel's enterprise-grade features such as thread-based state management, type safety, filters, telemetry, and extensive model and embedding support*."

Demo:
Create a New Agent (C#)

Demo:
Create a New Agent (Python)

Demo:
Semantic Kernel

Demo:

AutoGen

Demo:
Microsoft Agent Framework

Introduction to No-code Agent Development with Copilot Studio

Design and Implement Agents in Microsoft Copilot Studio

- Understanding Microsoft Copilot Studio

Poll: Which tool allows you to build and deploy AI agents without any coding skills?

- Azure AI Foundry
- Microsoft Copilot Studio
- AutoGen
- Semantic Kernel

Developing Agents with Microsoft Copilot Studio

- **Copilot Studio**
    a. Copilot Studio is a graphical, low-code tool for building AI agents!
    b. Can connect to other data sources (RAG)
    c. Copilot Studio
        i. Web App at https://copilotstudio.microsoft.com
        ii. Teams app

Developing Agents with Microsoft Copilot Studio

- **Copilot Studio**
    a. Copilot Studio is a graphical, low-code tool for building AI agents!
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        i. Web App at https://copilotstudio.microsoft.com
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Demo:

Create an AI Agent with Microsoft Copilot Studio

Conclusion

Resources

1. Azure AI Foundry
2. Upgrade from Azure OpenAI to Azure AI Foundry
3. Azure OpenAI in Azure AI Foundry Models quotas and limits
4. Add and configure models to Azure AI Foundry Models
5. What is Azure OpenAI in Azure AI Foundry Models?
6. Azure AI Foundry Playgrounds
7. What is Azure AI Foundry Agent Service?
8. Set up your environment (AI Foundry)
9. Quickstart: Create a new agent (Python, AI Foundry)
10. Quickstart: Create a new agent (C#, AI Foundry)
11. Models supported by Azure AI Foundry Agent Service
12. What are tools in Azure AI Foundry Agent Service?
13. Azure AI Foundry Agent Service quotas and limits
14. Azure AI Foundry pricing
15. Pricing calculator

Resources

16. What's new in Azure AI Foundry Agent Service
17. Threads, Runs, and Messages in Azure AI Foundry Agent Service
18. Built-in enterprise readiness with standard agent setup
19. Build collaborative, multi-agent systems with Connected Agents
20. Evaluate your AI agents locally with the Azure AI Evaluation SDK (preview)
21. Monitor Azure AI Foundry Agent Service
22. Develop an AI agent with Azure AI Foundry Agent Service (training module)
23. Call Azure Logic apps as functions using Azure OpenAI Assistants
24. Trigger an agent using Logic Apps
25. Azure OpenAI Assistants function calling
26. Azure AI Foundry SDK client libraries
27. Retrieval Augmented Generation (RAG) in Azure AI Search
28. Copilot Studio overview
29. Add knowledge to an agent
30. Semantic Kernel
31. AutoGen
32. Microsoft Agent Framework
33. AI agent adoption
34. Technology plan for AI agents

CERTIFICATION

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Azure Cookbook

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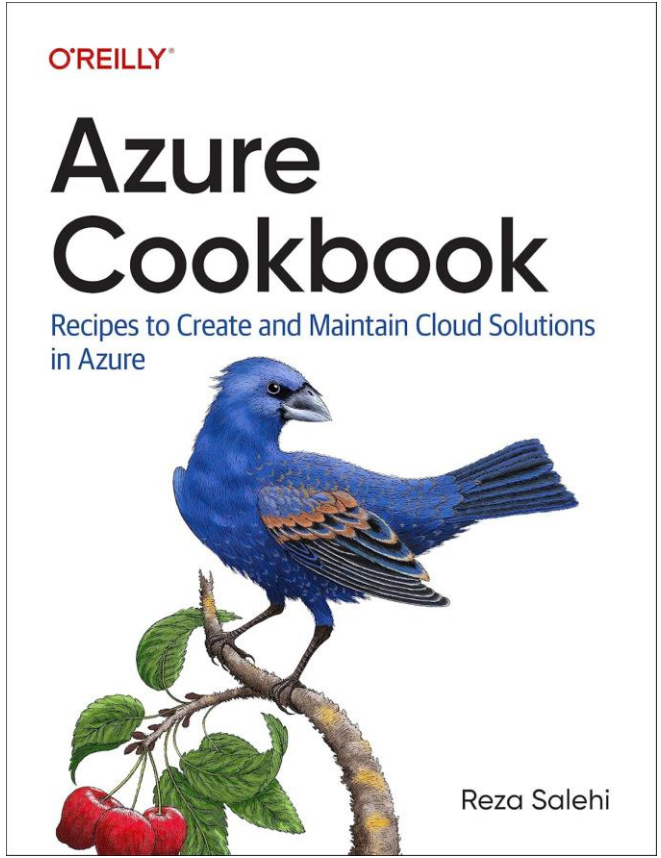

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